

Exploring the coherence and divergence between the objective and subjective measurement of streetscape perceptions at the neighborhood level: A case study in Shanghai

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Qiwei Song 

The Chinese University of Hong Kong, China

Yuxian Fang

Chongqing University, China

Meikang Li 

Shenzhen Technology University, China

Jeroen van Ameijde

The Chinese University of Hong Kong, China

Waishan Qiu 

The University of Hong Kong, China

Abstract

Understanding micro-level perceptions of street scenes is highly concerned with residents' behaviors and socioeconomic outcomes. While many studies rely on objective measures, such as physical features extracted from Street View Imagery (SVI) to proxy perceptions using derived formulas, others employ subjective measures from visual surveys to capture more subtle human perceptions. We argue that the two measurements can diverge significantly over the same perception concept, which might lead to opposite spatial implications in policy if not properly understood. Moreover, as perceptions are often examined individually, few studies have investigated their joint distribution patterns to reflect perceptions' multi-dimensional nature. To fill the gaps, we

Corresponding author:

Waishan Qiu, Department of Urban Planning and Design, The University of Hong Kong, 828A Knowles Building, Pokfulam Road, Hong Kong 000000, China.

Emails: qiuwaishan@126.com; waishanq@hku.hk

Data Availability Statement included at the end of the article

collected five pairwise perceptions from SVIs (i.e., complexity, enclosure, greenness, imageability, and walkability) at the neighborhood level in Shanghai. Each perception consists of pairwise values subjectively measured using a GeoAI-based approach and objectively quantified using formulas. We statistically and spatially compared the coherence and divergence of the two measures, further examining the perceptual differences. Advanced techniques including cluster analysis and factor analysis were employed to jointly evaluate their spatial distribution discrepancy. Our results revealed more differences than similarities between the two measures statistically and spatially, confirming any spatial implications concluded from one approach can vary significantly from the other. The joint spatial pattern further corroborated our conclusions. Our study enriches the literature on micro-level street perception measures, uncovers their critical differences to guide future comparative studies, and offers new approaches for urban perception mapping.

Keywords

Street view imagery, urban design quality, coherence and divergence, subjective versus objective perception, spatial heterogeneity

Introduction

Perceptions of street environment

Streets are the most emblematic open spaces embodying the essence of urban life and facilitating social interaction (Mehta, 2007; Montgomery, 1998). How residents perceive the micro-level street environment affects multiple behaviors and socioeconomic attributes, including walking, cycling, exercise, health and well-being, and housing premium (Almeida et al., 2021; Dong et al., 2023; Ito and Biljecki, 2021; Qiu et al., 2022; Song et al., 2024; Zhao et al., 2024). Independently from meso-level environment attributes (e.g., proximity to amenities), micro-level perceptions can significantly impact residents' activity and travel behavior (Ma and Cao, 2019). It is essential to establish an effective measurement for street perceptions, guide comparative studies, and offer practical insights into shaping human-centric street environments (Kim et al., 2014).

Despite its importance, consensus is lacking in street-scene perception measurement among scholars (Qiu et al., 2023). Some advocate objective measures through field data collection to define street physical characteristics, while others prefer subjective measures, gathering stated preferences through surveys and interviews (Lin and Moudon, 2010). Recently, the advent of Street View Imagery (SVI) has revolutionized both streams, enabling automated analysis of environment characteristics through a Geospatial Artificial Intelligence (GeoAI) approach (Chen and Biljecki, 2023; Ma et al., 2021).

Analyzing SVI with Machine Learning (ML) has enabled high-throughput measures of objective streetscape elements (e.g., trees and sky) through semantic segmentation (Gong et al., 2018), quantifying their view indices (i.e., proportion in the image) to represent street qualities. On top of this, design perceptions (e.g., enclosure, imageability) were further quantified by recombining relevant view indices through formulas derived from theories and expert knowledge (Ma et al., 2021). For instance, imageability can be proxied by aggregating view indices of buildings, signs, and skyscrapers, which describes street richness and diversity according to its definition in urban design theory. In this paper, we refer to Ma et al. (2021)'s approach as the "objective measure" of perceptions. Despite being widely applied (He et al., 2023; Rui and Xu, 2024; Wang et al., 2022), such rule-based constructs of perceptions can be arbitrary (Qiu et al., 2023). In contrast, in this

paper, subjective measure refers to the perceptions predicted with ML algorithms trained on street-scene preferences from crowd-sourced visual surveys (Salesses et al., 2013) or expert panels (Qiu et al., 2022). Integrating online surveys, SVI, and ML overcome the limitation in sample size associated with the conventional surveys (Dubey et al., 2016).

Although the selection of objective or subjective measures may primarily depend on data availability, a clear understanding of how these measures overlap and differ is needed to improve the reliability of their outcomes.

Divergence in the two measures

As most cities are socioeconomically heterogeneous, street qualities vary across different neighborhoods (Sallis et al., 2011; Zhou et al., 2019), whichever measure is employed. Significant spatial disparities were documented regarding visual greenery (Li et al., 2016; Wang et al., 2021) and walkability (Su et al., 2019; Zhou et al., 2019), implying substantial social inequalities. Perception mapping can deepen our understanding of such issues and support informed interventions for management (He et al., 2023). However, using different measures for perceptions might yield severe discrepancies in their outcomes.

Objectively measured perceptions (such as enclosure, greenness, and complexity) have been found to significantly impact cycling (Lu et al., 2019), children's walking to schools (Wang et al., 2022), and the elderly's walking (Wei et al., 2024). These objective measures have been widely adopted in urban studies (He et al., 2023; Zhou et al., 2019). On the other hand, subjectively measured perceptions have revealed a more robust relationship with housing prices (Qiu et al., 2022) or crimes around subway entrances (Su et al., 2023) compared to objective features such as streetlights. Furthermore, when measuring the same perception concept, subjective measures may outperform their objective counterparts, for example, in explaining housing prices (Qiu et al., 2023). Moreover, neighborhoods perceived as lacking enclosure using objective measures might be viewed as highly enclosed using subjective measures because of their underlying psychological process (Ramírez et al., 2021).

This discrepancy raises critical questions: Do the two measures interpret the same perception concept differently? If the two measures differ, how can policymakers apply them interchangeably to inform decisions?

This points to the first research gap. As scholars and planners increasingly depend on either way to draw spatial implications (such as social equality and walkability), biased conclusions or unintended consequences might come without adequately considering the divergence between the two different approaches (Qiu et al., 2023) over an identical perception concept. For example, different statistical patterns and spatial heterogeneity by objective and subjective measures may even yield contrary policy implications. Few studies have systematically examined the "perceptual difference," which refers to the disparities between the two measures in spatial mapping (Kang et al., 2023).

Revealing perception patterns comprehensively

Furthermore, people barely assess the environment using a single perception or weigh perceptions equally. Different perceptions are often mapped separately for spatial interpretations (Wu et al., 2023). However, not only is the physical environment interwoven into perceptions (Ramírez et al., 2021), but different perceptions are also intercorrelated (Ewing and Handy, 2009). Despite its multi-dimensional nature, few studies have examined perceptions synergistically to describe environmental characteristics (Liang et al., 2023; Ma et al., 2021).

This implies the second research gap in exploring an integrated evaluation framework to comprehensively comprehend perceptions' spatial heterogeneity. For instance, unsupervised ML,

including cluster analysis, offers a new perspective on uncovering various multi-perception profiles across measures (Liang et al., 2023; Su et al., 2019). Additionally, multi-collinearity among perceptions is another issue often arising in analyses (Zhang et al., 2018). Tools such as Factor Analysis (FA) can address this and reveal underlying latent attributes to retain their core factors for further comparison (Kang et al., 2021).

Research objectives

To fill the gaps, this study provides an in-depth understanding of the similarities and differences between two prevalent measures for micro-level design perceptions to identify potential biases in environmental assessments. Our main contributions are two-fold: (1) statistically and spatially comparing the coherence and divergence of the two measures (i.e., subjective perception predicted using GeoAI approach; objective perception quantified from formulas); and (2) systematically and comprehensively examine the joint spatial patterns of perceptions, offering new ways for urban perception mapping.

Literature review

Objectively measured perception

As SVI is increasingly used to collect large-scale street environment datasets (Ito and Biljecki, 2021), computer vision (CV) is employed to classify visual elements objectively. Researchers have quantified visual greenery (Li et al., 2016; Lu et al., 2019), sky, and building proportions to categorize urban scenes (Gong et al., 2018). Others have developed rule-based processes to determine design perceptions objectively using various features. For example, Zhou et al. (2019) proxied the visual walkability index from several sub-indicators, including greenery, crowdedness, enclosure, and pavement. Notably, Ma et al. (2021) compiled greenness, openness, enclosure, walkability, and imageability using definitions from design theories (Ewing and Handy, 2009; Lynch, 1960).

Research has proved the effectiveness of objective measures (Wei et al., 2024). Particularly, He et al. (2023) found that integrating design perceptions (e.g., imageability and greenness) can improve the accuracy of urban renewal assessments. Wang et al. (2022) reported that perceptions such as enclosure positively correlated with the odds of children walking to school regardless of their background, implying the importance of a visually defined environment. These analyses imply that objective measures of perceptions can be essential for researchers to reflect street characteristics.

Subjective measures

Subjective measures traditionally relied on opinions collected from surveys or expert panels (Ewing and Handy, 2009) and have recently incorporated SVI to conduct large-scale online surveys on street perceptions (Chen and Biljecki, 2023). The MIT Place Pulse project collected massive crowd-sourced data (Dubey et al., 2016; Salesses et al., 2013), which was utilized to predict city-wide psychological perceptions (e.g., lively) using high and low-level features such as street elements or color balance (Zhang et al., 2018). More recent studies have measured design perceptions following this framework. Particularly, Qiu et al. (2022) quantified complexity, enclosure, imageability, human scale, and safety, using experts' inputs on visual surveys to train ML algorithms for perception mapping. It further revealed that imageability, complexity, and safety positively impacted housing prices, whereas enclosure and human scale suggested adverse effects. Su et al. (2023)

also found that subjectively measured imageability and human scale were related to lower crime rates around subway station entrances. These studies suggest that subjectively measured perceptions can also inform socioeconomic dynamics at the city scale.

Subjective vs. objective measures

Nevertheless, given the effectiveness of both measures in the literature, few studies have examined the statistical and spatial divergence between them regarding the same design perceptions (Qiu et al., 2023). A place would likely diverge in its qualities when comparing across measures, which could significantly affect planning implications if perception mapping is used to monitor long-term street characteristics (Ye et al., 2019). Specifically, Qiu et al. (2023) found that the objective measure of greenness outperforms the subjective counterpart in explaining housing price variations. In this regard, whether the two measures are similar enough to be used for analysis interchangeably is still unknown. Such a knowledge gap warrants us to compare the measures both statistically and spatially systematically. Moreover, the multi-perception analysis can offer distinct insights compared to a univariate perception analysis (Su et al., 2019).

Data and methods

Analytical framework and study area

Our case study focused on areas within the Outer Ring Road of Shanghai. Shanghai is a critical hub for finance, trade, and shipping in China. Using both measures across neighborhoods, our city-wide perception mapping can reveal important implications for monitoring the street environment to implement the ongoing renewal action plan.

The analytical framework is sevenfold (Figure 1). First, we gathered SVI, employing CV to extract indices of over 30 physical features. Second, for subjective perceptions, we collected experts' inputs through an online visual survey for sample SVIs and chose the best-performed ML models for prediction. The predicted subjective scores were further validated using opinions from non-experts. Third, we calculated the objective counterparts of the same perception concepts based on an established framework (Ma et al., 2021). Fourth, we conducted Pearson correlation and paired t-tests to scrutinize the differences between the two measures. Fifth, we mapped perceptions to identify their spatial heterogeneity and analyzed the "perceptual difference." Sixth, perceptions were clustered to jointly analyze their overlaps and discrepancies. Lastly, we reduced the perception dimensionality using Principal Component Analysis (PCA) to construct new factors for FA. We examined the new variables of both measures to compare their spatial discrepancy.

Selection of five perceptions

Following established theoretical frameworks (Ewing and Handy, 2009; Lynch, 1960; Nasar, 1987), we included five design perceptions to reflect eye-level experience, namely, "complexity," "enclosure," "greenness," "imageability," and "walkability" (see Supplement Table S1 for detailed definitions and measures). Three perceptions (complexity, enclosure, and imageability) were operationalized by Ewing and Handy (2009), laying the foundation for subjective measurement. Ma et al. (2021) objectively measured four perceptions by recombining view indices, providing a high-resolution measurement paradigm. Visual greenery (greenness) is essential for the urban built environment and planning (Li et al., 2016; Lu et al., 2019), while walkability is instrumental in building a healthy city (Zhou et al., 2019).

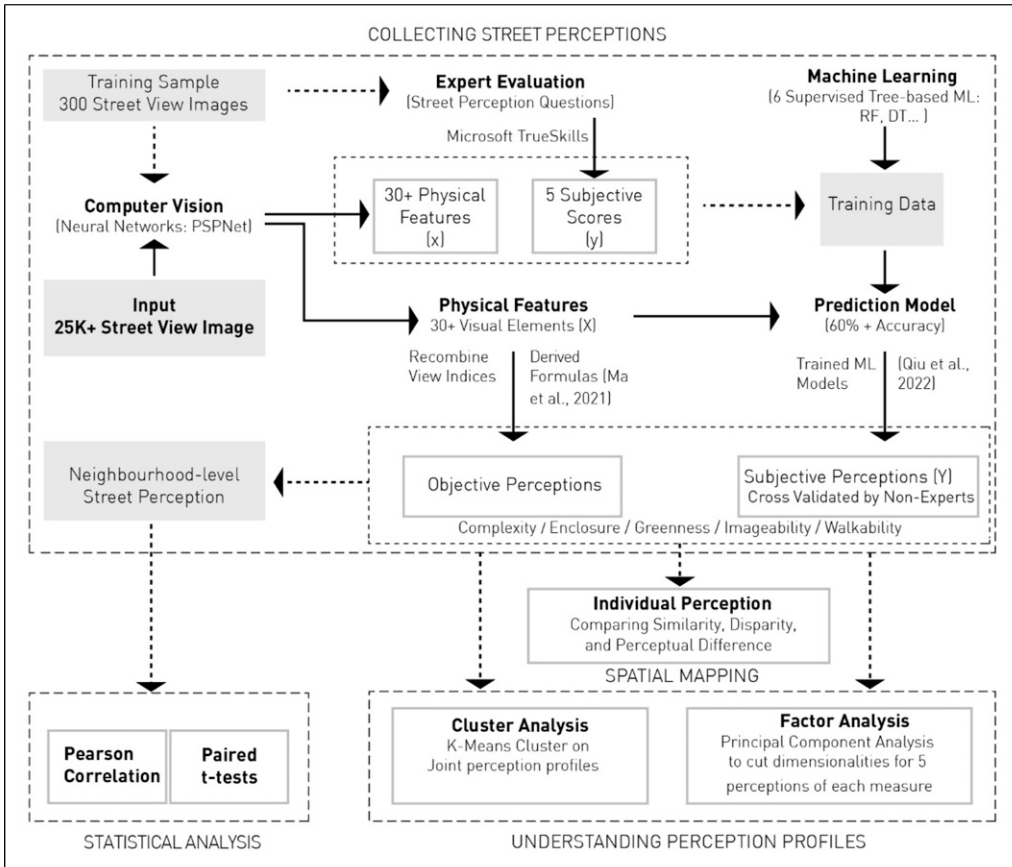


Figure 1. Analytical framework.

Calculating subjective perceptions

Sampling and downloading the SVI. We sampled SVI at 50m intervals (Ma et al., 2021) along the public street center lines using QGIS. The center line shapefile was from Open Street Map (<https://www.openstreetmap.org/>). As the typical block width/length in Shanghai ranges between 300–500m (NYU, UN Habitat and Lincoln Institute, 2016), a 50m interval guarantees that approximately 6–10 images can be sampled along each street. The SVI was downloaded by feeding the coordinates into the Baidu Street View API. The detailed SVI request settings are indicated in the Supplementary Material. Following data cleansing to eliminate interior views, we retrieved 25,276 valid SVIs.

Online visual survey collecting local subjective perceptions. To ensure broad geographical coverage of varied urban landscapes, 300 SVIs were randomly sampled to capture diverse streetscapes in Shanghai for the visual survey, of which more samples fell within Outer Ring Road (see Supplementary Figure S1). None of these 300 SVIs contained blurred, dark, or interior images (Ito and Biljecki, 2021). A website was developed where participants were shown two randomly paired SVIs. They were asked to choose the preferred image to answer the perception assessment questions. Questions were designed as straightforward as “Which place has better complexity?”

with the definition of complexity indicated for guidance (see [Figure S2](#) in the Supplementary Material).

In contrast to research that elicited opinions from the general public ([Dubey et al., 2016](#); [Salesses et al., 2013](#)), we collected scores from expert panels, which is advantageous for evaluating design-related perceptions of which the public is not well-versed ([Ewing and Handy, 2009](#); [Ye et al., 2019](#)). Methods in the Supplementary Material include details regarding our recruitment of an expert panel and additional considerations that helped stabilize the trustworthy pairwise comparison results. In short, we collected an average of 29.5 pairwise votes for each SVI, deemed appropriate compared to previous research ([Dubey et al., 2016](#); [Zhou et al., 2022](#)). The clicks were converted into a continuous score using the TrueSkill algorithm (see Methods section in the [Supplemental Material](#)).

Semantic segmentation to exact view indices. Utilizing Pyramid Scene Parsing Network (PSPNet), a widely adopted semantic segmentation algorithm ([Zhao et al., 2017](#)), we computed view indices of over 30 physical features from SVI (see Supplemental Methods section). [Figure S3](#) in the Supplement illustrates examples of segmentation results. As shown in Supplement [Table S2](#), previous literature ([Ewing and Handy, 2009](#)) has proved that at least 10 components (such as sky, trees, roads, and buildings) significantly impact human perceptions.

Prediction of subjective perceptions. Using view indices as the explanatory variables ([Qiu et al., 2022](#)), we chose five ML algorithms, training them (80% for training, 20% for testing) to predict subjective perceptions of the 300 SVI rated through visual surveys. The performance of ML models can be viewed in Supplement [Table S3](#), with its explanations elaborated in the Methods section in the Supplementary Material. To summarize, Gradient Boosting performed well in predicting four qualities, such as greenness, while Random Forest outperformed other algorithms in predicting complexity. The model's prediction accuracy was acceptable, given the size of the training set, and we further normalized the results to a 0–1 scale (worst to best). Using the best-performed algorithms, we predicted subjective perceptions for all SVI.

Validating subjective perceptions using non-experts' input. Meanwhile, we conducted manual validation to assess how experts' opinions align with the general public's perceptions. We recruited 24 non-experts (no design background) and explained each perception to them. To ensure the participants' demographics were balanced, we included three male and three female participants from the following age groups: 20s, 30s, 40s, and 50s. They were asked to compare five representative SVIs, ranking each perception high to low.

Kendall's W was applied to check the inter-rater reliability of perception ranking. The results indicated that non-expert raters achieved substantial to high levels (Kendall's $W = 0.5$ to 0.8) of agreement on ranking straightforward perceptions, including greenness, complexity, and enclosure. Notably, it demonstrated lower consensus on imageability and walkability, with Kendall's W suggesting modest agreement. It generally obtained good inter-rater reliability, while more nuanced concepts might require specific design expertise for more consistent evaluation.

Based on the non-experts' rankings, we calculated the average ranking for SVI as “non-experts' consensus ranking.” We then compared these rankings with the predicted perception scores generated by ML algorithms, using the Jaro–Winkler similarity score to assess the similarity of their order by converting the ID into characters. The result revealed good coherence between the predicted and non-experts' rankings, with most perceptions scoring between 0.93 and 1, except for imageability, which scored 0.78. It indicates a strong consensus between the predicted and non-experts' rankings of SVI. The validation confirmed that the subjective measure broadly reflects the general public's understanding of urban scenes. The subjective perception scores of SVI ([Figure S4](#)) and validation metrics ([Table S4](#)) are detailed in the Supplementary Material.

Calculating objective perceptions

To measure objective perceptions, we followed [Ma et al. \(2021\)](#) and [Qiu et al. \(2023\)](#), using formulas to recombine important streetscape view indices (e.g., sky, tree, sidewalk) based on operational definitions (see Supplement [Table S1](#)). These scores were then standardized on a scale of 0–1 (worst to best) for ease of comparison.

Discovering the coherence and divergence of perception mapping

Revealing statistical differences. To confirm the statistical coherence and divergence, we conducted the Pearson correlation between the two measures for each perception. Furthermore, paired t-tests were employed between subjective and objective perceptions, which confirms whether they are significantly different regarding data distribution.

Mapping perceptual differences. Prior studies have shown that streetscape perceptibility is better evaluated by zone rather than grid ([Su et al., 2019](#)), which also applies to perceptions. The SVI was aggregated to 4604 Areas of Interest (AOI) units for further comparison, and each AOI contains approximately 5.5 SVI. The average size of AOIs is 0.056 square kilometers, with a mean population of 4982 in each geographical unit. We visualized the spatial patterns of the subjective and objective perceptions using natural breaks, accounting for their overlaps and divergences. To explicitly capture the differences between subjective and objective measure, inspired by previous literature ([Kang et al., 2023](#)), we defined and mapped the perceptual difference as the subjective perception minus the objective perception. Following prior studies ([Zhou et al., 2024](#)), we calculated the “between variance” for both perception measures, reflecting perception heterogeneity at the AOI level.

Cluster analysis of perceptions. Notably, to enable a more integral evaluation, perceptions were analyzed in groups to reveal spatial heterogeneity ([Ma et al., 2021](#)). Cluster analysis was employed to categorize subjective and objective perceptions and uncover various perception profiles. In this research, we employed the K-means algorithm, an unsupervised ML technique known for its simplicity and effectiveness. This method efficiently minimizes the distances between points and their cluster centroids, optimizing within-cluster cohesion ([Jain, 2010](#)). Inspired by prior research ([Su et al., 2019](#)), we adapted the clustering procedure in three steps: (1) Determining the optimum cluster number k using the elbow method ([Liang et al., 2023](#)) in K-means and validating the result using the silhouette coefficient, resulting in $k = 4$ as the optimum number; (2) Applying K-means cluster analysis to cluster scores into four groups for both measures; (3) Conducting one-way analysis of variance (ANOVA) to compare the mean values of each perception, labeling the four clusters of areas by four classes of quality (i.e., good, medium, low, very low). Such analysis can successfully reject the null hypothesis that observed differences among different groups are due to random variation. Moreover, these delineated distinct clusters were mapped spatially to reveal their nuanced coherence and divergence across measures.

PCA perceptions using both measures. To reduce the dimensionality of perceptions, we conducted FA using PCA, which can address multi-collinearity among variables ([Kang et al., 2021](#)). This approach yielded two new orthometric variables for subjective and three for objective perceptions. We further examined the distribution patterns of these introduced factors to gain more insights into their concordance and discordance.

Results

Descriptive statistics of perceptions

Supplement [Table S5](#) displays the descriptive statistics of subjectively and objectively measured perceptions. Interestingly, we found that all subjective scores exhibit higher mean values than their objective counterparts. This suggests that other factors, such as visual appearances and emotions, could also contribute to subjective perceptions ([Kang et al., 2021](#)).

Statistical differences between measures

[Table 1](#) indicates that for most perceptions (i.e., imageability, walkability, complexity), subjective measures reported negligible correlation with their objective counterparts. Nevertheless, more similarities can be detected in greenness and enclosure, which showed moderate and low correlations.

The *p*-values of the paired *t*-tests are all less than 0.01, revealing significant differences in measuring the same concept. Notably, *t*-statistics of Imageability (*t* = 35.12) and walkability (*t* = 76.71) show lower magnitude compared with other qualities, with greenness (*t* = 337.88) demonstrating the most significant deviation. It implies that objective measures of imageability and walkability can better proxy the subjective nuances using relevant street features. Meanwhile, greenness is more complicated regarding how people sense the place subjectively ([Wang et al., 2021](#)).

Spatial heterogeneity of individual perceptions

[Figure 2](#) illustrates the distribution of subjective and objective measures and perceptual differences for the five perceptions within the Outer Ring Road in Shanghai. In general, enclosure, greenness, and walkability display similar distribution patterns when identifying good-quality areas using both measures. However, complexity and imageability exhibit significant disparities and even opposite patterns.

A strong consensus is found in identifying very low or low-quality areas regarding greenness. Nevertheless, using subjective measures, the site was interpreted greener than objective measures. Notably, most north Pudong areas are classified as good or medium quality using subjective measures, while objective perception rates them as medium or low. The junction of Jing An, Zha Bei, Pu Tuo, and Hong Kou districts exhibits the most prominent differences, as supported by the perceptual difference mapping.

Regarding walkability and enclosure, higher subjective perceptions concentrate in sub-city centers in Lu Wan, Xu Hui, Zha Bei, Jing' An, Hong Kou, and downtown Pu Dong. Objective

Table 1. Pearson Correlation and Paired *t*-tests Results.

Pearson correlation analysis and Paired <i>t</i> test variables		Pearson correlation coefficient <i>r</i>	<i>t</i> -score
S1_CMPLX & O1_CMPLX	Complexity	−0.070***	242.202***
S2_ENCLS & O2_ENCLS	Enclosure	0.445***	76.714***
S3_GREEN & O3_GREEN	Greenness	0.703***	337.888***
S4_IMGBL & O4_IMGBL	Imageability	0.115***	35.128***
S5_WALKB & O5_WALKB	Walkability	−0.007	57.846***

Note: Significance level ****p* < .01.

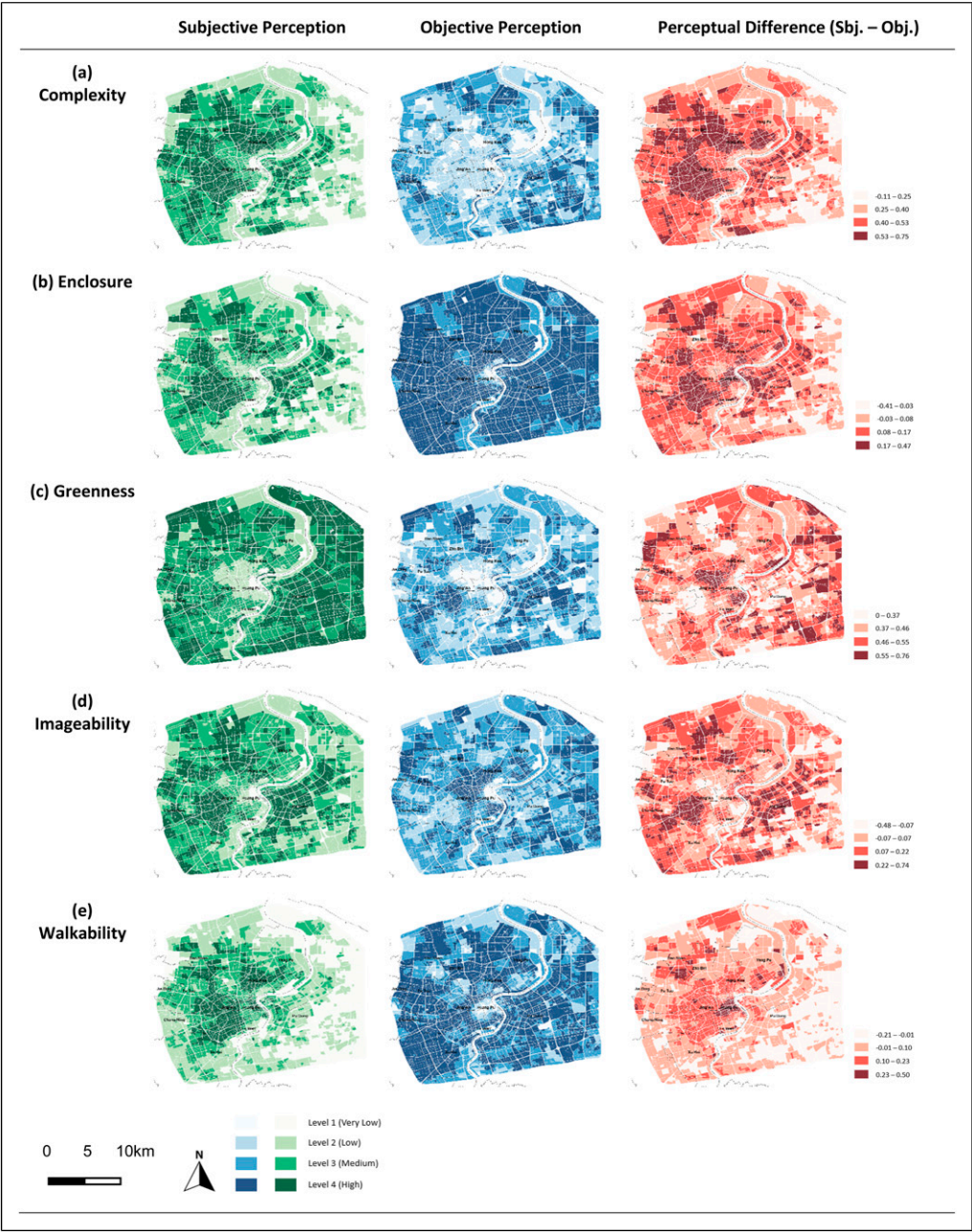


Figure 2. Mapping subjective, objective perceptions, and perceptual differences. (a) Complexity. (b) Enclosure. (c) Greenness. (d) Imageability. (e) Walkability.

perceptions, however, are more equally distributed across all districts, indicating more areas with high walkability and enclosure.

Regarding complexity, the objective measure rates most land in Jing' An, Lu Wan, Huang Pu, north Xu Hui, central Hong Kou, and downtown Pu Dong as low quality, contrary to the subjective

perception. In contrast, when comparing the two measures, the reverse pattern applies to imageability around the intersection of Xu Hui, Lu Wan, Chang Ning, and Jing' An. The Eastern bank of Huangpu River, the financial district in Pu Dong, also exhibits marked differences. Perceptual difference maps further corroborate the discrepancies.

Moreover, beyond investigating visual patterns, the between variance metric (see [Table S6](#) in the Supplement) provided more clues on their spatial heterogeneity in general. It reveals that subjective perceptions of complexity exhibit more significant variance than objective measures, indicating higher spatial heterogeneity among AOIs. Conversely, objective greenness displays greater variance than its more homogeneous subjective counterpart. We observe a higher variance in subjective enclosure and objective imageability, while the variance of subjective walkability is only marginally higher than that of the objective measure. These findings align with the general distribution in spatial patterns, although the specific spatial pattern of perceptions may diverge, as visualized in the perceptual difference maps.

Clustered perceptions

[Figure 3](#) illustrates distinct perception profiles categorized by K-Means clustering analysis. The average perceptions for each cluster are detailed in [Table S7](#) in the Supplementary Material. Examining the spatial heterogeneity between the two measures offers a comprehensive interpretation of street perceptions. Notably, both measures divided the areas into four groups. ANOVA tests confirmed that all F-statistics are significant at the 1% confidence level, indicating that the clusters significantly differ, supporting the method used.

Specifically, Group 4 from the subjective measure is characterized by poor complexity, greenness, and imageability, with low enclosure. It is mainly concentrated in the Huang Pu district on the west riverbank. Interestingly, these areas are defined as Group 3 using objective measures, representing the worst joint perceptions. Group 2, in the subjective measure, implies good greenness but low complexity, imageability, enclosure, and very low walkability. Similarly, Objective Perception Group 2 has low-quality values in all sub-qualities except for imageability. At the intersection of Jing' An, Lu Wan, Zha Bei, Pu Tuo, and Huang Pu districts, it is primarily occupied by Objective Group 2. Nevertheless, using subjective measures, they are categorized as Group 3, featured by medium complexity, imageability, and enclosure, with low walkability and greenness.

Group 1, using subjective measures, exhibits good complexity, enclosure, and imageability, with medium-quality greenness and walkability. It encompasses downtown Pu Dong and extends across the west side of the Huang Pu River, covering Jing' An, Lu Wan, Zha Bei, Yang Pu, Hong Kou, and Pu Tuo. Similarly, in the objective measure, Group 1 shows good complexity, enclosure, greenness, and walkability but low imageability. It remains the highest quality group objectively despite being more scattered and not matching the geographic extent of its subjective counterpart. The differences between the groups are partly explained by objectively measured Group 4, which has medium quality in all five perceptions.

Factor analysis and mapping

[Figure S7](#) in the Supplementary Material indicates that some perceptions exhibited potential collinearity, particularly in subjective measures (also see Methods section in the Supplement). Therefore, we applied PCA to all five perceptions to reduce the dimensionality for subjective and objective measures. Following previous research ([Kang et al., 2021](#); [Wang et al., 2021](#)), the correlation matrix was generated, and we calculated eigenvectors/ eigenvalues. Following the Kaiser-According to the Guttman rule, we chose principal components whose eigenvalues ranked top. We selected the top two principal components (explaining variance ratio of 0.627 and 0.307,

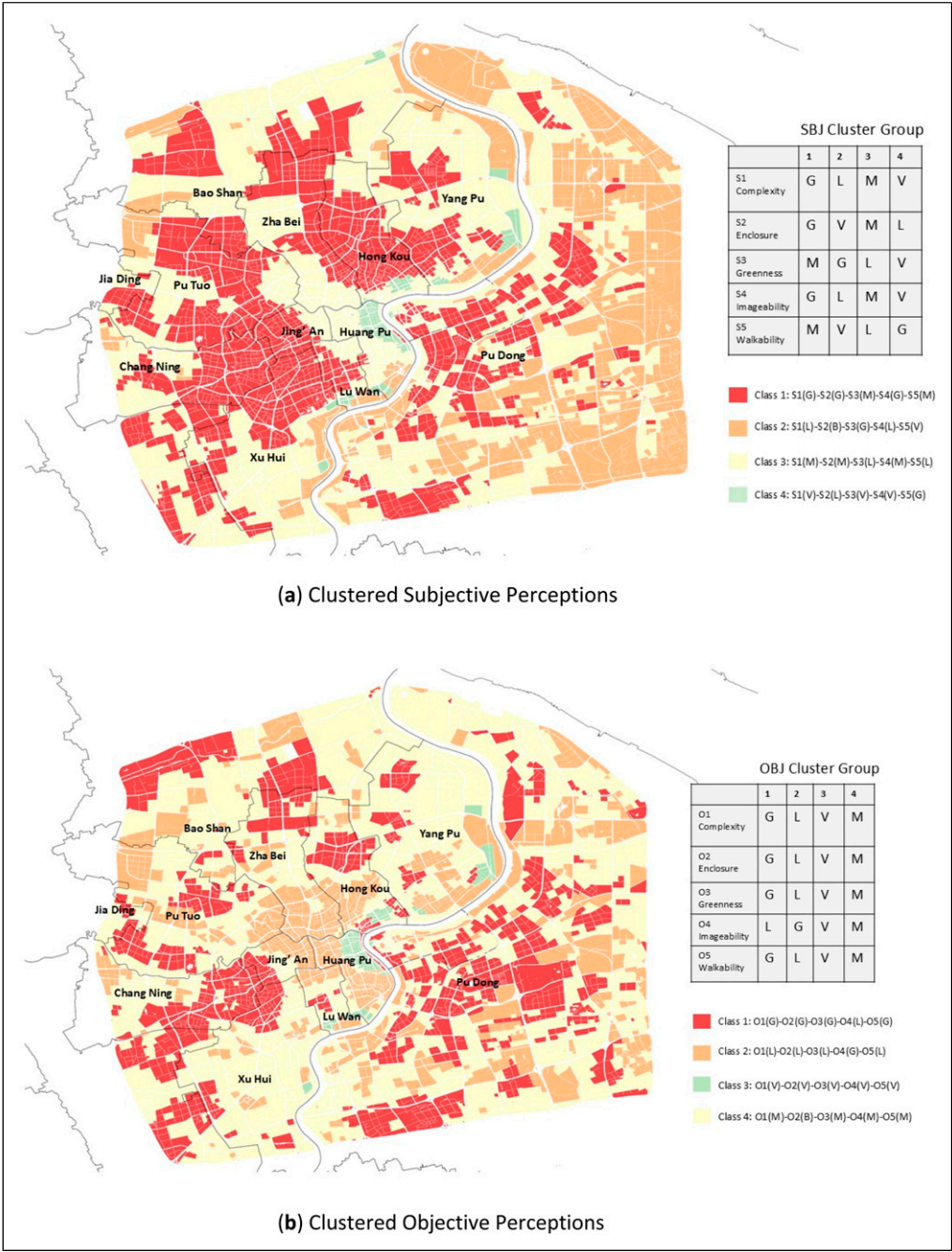


Figure 3. Perception mapping (a) subjective and (b) objective perceptions. (Quality: G = good, m = medium, L = low, V = very low).

respectively) as new factors for the subjective measure, which substantially explained 93.4% of total variances in original perception variables, indicating only less than 7% information loss in original variables kept in the rest of the components. Similarly, for the objective measure, we selected the top three principal components (explaining variance ratio of 0.436, 0.297, and 0.149, respectively),

which explained 88.2% of the total variances. The outcomes explained better total variances than Wang et al. (2021) (i.e., 86%) and were partially better than Kang et al. (2021) (i.e., 84%, 88%, 94%, and 98%). Table 2 lists the factor loadings of the variables, which suggests the correlation coefficients between the new factors and the original perceptions. For instance, OBJ Factor 3 revealed a strong positive relationship (coef. = 0.863) with complexity. Following prior cut-off thresholds, variables with absolute values over 0.30 were deemed proper for interpreting new factors (Kang et al., 2021).

Using the subjective measure, SBJ Factor 1 shows negative loadings on complexity, enclosure, imageability, and walkability. Therefore, factor 1 can be best described as “Suburban.” This indicates that the higher the score, the more suburban it feels. SBJ factor 2 reveals negative loading on greenness but high positive loading on walkability, which can be reinterpreted as “Urban,” as it stresses visual walkability but typically contains lower greenness. However, the street profiles are narrower than the main streets, rendering a human-scale experience.

Using the objective measure, OBJ Factor 1 reveals negative loadings on complexity, enclosure, greenness, and walkability. Therefore, it can be similarly interpreted as “Suburban.” The following principal component (OBJ Factor (2) leans toward the feeling of greenness (positively), imageability (negatively), and complexity (positively). This represents the degree of “Natural.” OBJ Factor 3 positively contributes to complexity and negatively correlates with walkability and can be interpreted as the degree of “Urban.” It is worth mentioning that though both measures we have reinterpreted perceptions of “suburban” and “urban,” they represent different sub-quality profiles; in other words, the subjectively measured “urban” (SBJ Factor (2) and objectively measured “urban” (OBJ Factor (3) do not denote the same characteristics.

Figure 4 illustrates the distribution of the new factors. The mapping of SBJ Factor 1 indicates that the Huang Pu River presents the highest level of “suburban.” In contrast, the downtown Pu Dong and the intersection of Hong Kou, Huang Pu, Jing’ An, Zha Bei, and Pu Tuo districts are

Table 2. Factor Loadings of Perceptions using PCA.

		SBj factor 1	SBj factor 2	
Subjectively measured scores		Suburban	Urban	
S1_CMPLX	Complexity	−0.544	0.074*	
S2_ENCLS	Enclosure	−0.532	0.124*	
S3_GREEN	Greenness	−0.226*	−0.713	
S4_IMGBL	Imageability	−0.507	−0.297*	
S5_WALKB	Walkability	−0.334	0.617	
		Obj factor 1	Obj factor 2	Obj factor 3
Objectively measured scores		Suburban	Natural	Urban
O1_CMPLX	Complexity	−0.367	0.303	0.863
O2_ENCLS	Enclosure	−0.606	−0.210*	−0.186*
O3_GREEN	Greenness	−0.368	0.602	−0.246*
O4_IMGBL	Imageability	−0.283*	−0.707	0.221*
O5_WALKB	Walkability	−0.529	−0.010*	−0.333

Note: * denotes the absolute value <0.30, which is considered no significant correlation and is excluded from explaining new factors.

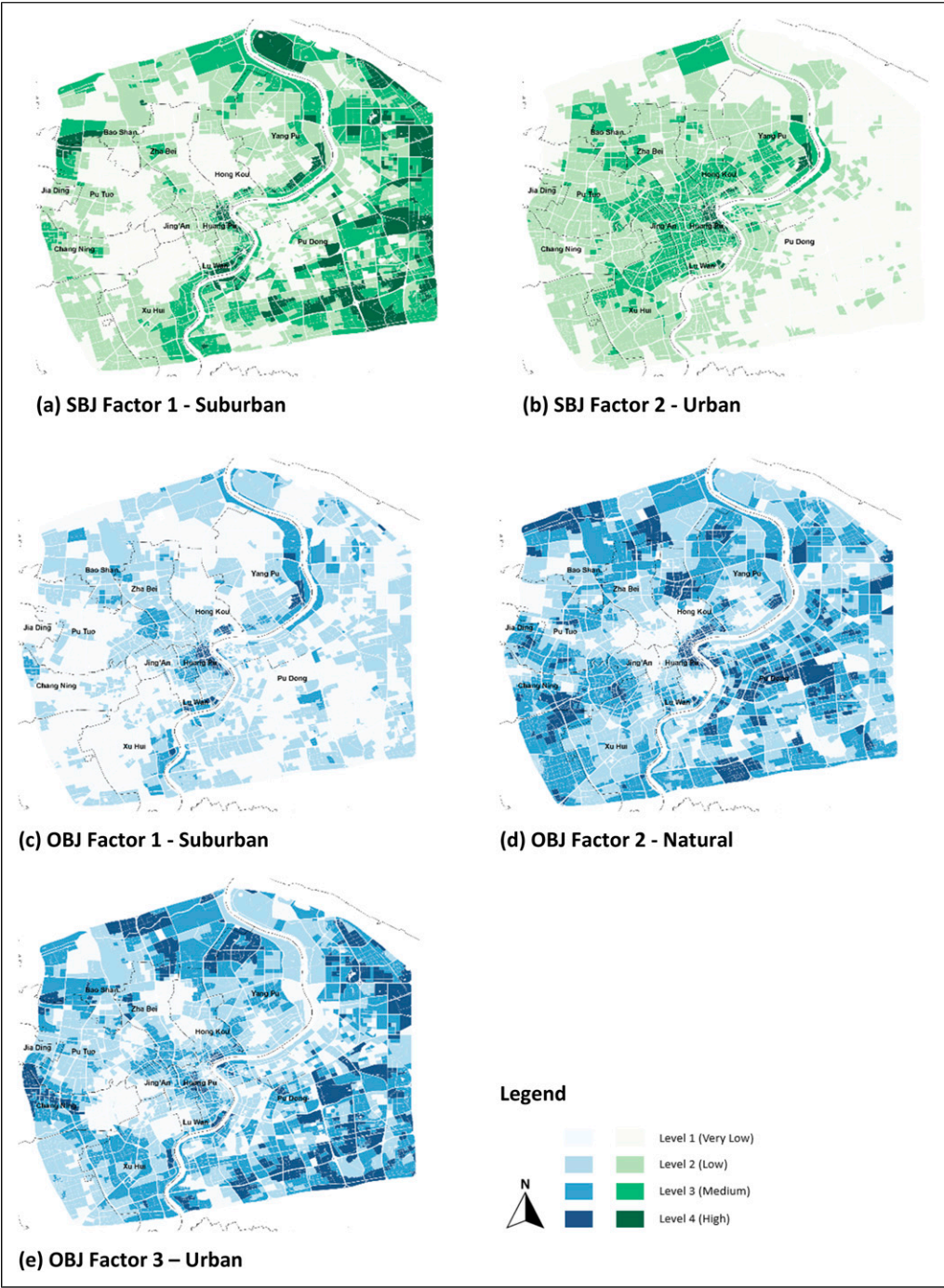


Figure 4. Distribution of reinterpreted perceptions (a) SBJ Factor 1, (b) Factor 2 for subjective measures, and (c) OBJ Factor 1, (d) Factor 2, and (e) Factor 3 for objective measures.

characterized by an “urban” (SBJ Factor (2) experience. This sense of urbanity gradually fades away from the center.

For objective measures, OBJ Factor 1 (Suburban) exhibits a similar pattern in identifying the highest score areas as to SBJ Factor 1. Moreover, unlike SBJ Factor 1, more areas are identified in the level 1 and 2 score ranges. Higher scores (level 3 and level (4) of OBJ Factor 2 (Natural) mainly concentrate in the intersection of Hong Kou and Huang Pu districts and scatter around eastern downtown Pu Dong. Lower score ranges (Level 1 and Level (2) of OBJ Factor 3 (Urban) can be detected at the intersection of Jing 'An, Lu Wan, and Chang Ning districts and the junction of Zha Bei and Hong Kou districts. These findings are further cross-validated by examining representative SVI of the new variables (see Supplement [Figure S](#)).

Discussion

Mismatch of perceptions between two measures

This research brings a fresh perspective to the coherence and divergence between the two perception measurement frameworks, both statistically and spatially. First, Pearson correlation and paired t-tests confirm our hypothesis that the two measures of an identical perception are significantly different. Notably, the mean values of all the subjective perceptions surpass their objective counterparts. For instance, when comparing the histogram of greenness, the subjective measure concentrates on a higher but narrower score range. This suggests that the subjective measure relies on the overall experience of the visual scenes ([Ramírez et al., 2021](#)), and its inherent complexity has yet to be captured in the objective measure. As previous scholars noted, for less straightforward perceptual qualities, simply aggregating or recombining visual features may result in information loss ([Qiu et al., 2023](#)).

Second, perception mapping suggests that street perceptions are distributed unevenly, with subjective perceptions revealing more robust heterogeneity patterns. Urban and sub-city centers (e.g., Jing' An and Xu Hui districts) show higher complexity, enclosure, imageability, and walkability but lower greenness. It aligns with our intuition that in such densely populated districts, the street environment and interfaces are more complex, containing more residential towers framing more enclosure, memorable compared to suburban developments, and more walking-friendly due to their smaller block sizes. However, narrow street profiles and limited spaces can cause less street greenery in these neighborhoods.

Lastly, when making pairwise comparisons between measures for identical perceptions, the between variance metric further supports the general trend of heterogeneity patterns, with subjective complexity and objective greenness showing more substantial heterogeneity than their counterparts. Notably, the two measures exhibit a similar variance in walkability among AOIs. Though the general heterogeneity trend can be detected using the metric, the perceptual difference map allows for a more explicit interpretation of where the two measures may spatially exhibit high divergence. Enclosure, greenness, and walkability show comparable patterns across measures while showing drastic discordance in complexity and imageability. For instance, using an objective measure, most land in downtown Pu Dong is less complex, while the subjective method draws a contrary conclusion.

Investigating mismatch of perceptions jointly

We further generated the street perception profiles for all communities by the clustering approaches ([Liang et al., 2023](#); [Su et al., 2019](#)). The cluster analysis of multi-perceptions provides an integrated

perspective when assessing street quality profiles by classifying perceptions into four groups for both measures.

We conclude that the two methods consistently identify areas of very low quality, while the subjective measure reveals more jointly best-quality areas than the objective measure. Group 4, using the subjective measure, characterizes areas in Huang Pu district and West Riverbank, with Group 3, using the objective measure, representing the lowest joint quality profile. Both Group 1 represent best-quality clusters; objective Group 1 has four good qualities except low imageability, while subjective Group 1 shows three good qualities except medium greenness and walkability.

Furthermore, spatial heterogeneity is found jointly between medium- and low-quality groups. We identified similarities in spatial distribution across measures around the Pu Dong district, while discrepancies exist west of the riverbank. Objective Group 1 is more dispersed than its counterpart. Their disparity in individual perceptions may explain this joint spatial divergence.

Additionally, Factor Analysis effectively reinterprets perceptions and explicitly uncovers latent characteristics of perceptions that can fuse various sub-qualities using different measures. It enriches existing literature by introducing new perceptions, including the sense of “Urban,” “Suburban,” and “Natural” through measured urban design qualities. These constructed dimensions offer valuable insights into understanding the general city image, adding to the classical design literature (Lynch, 1960; Montgomery, 1998).

Implications

This study presents some coherence but more divergence between these two measures statistically and spatially, which may be explained by the difference in underlying mechanisms related to the psychological factors that objective measures have not incorporated. Our study provides implications for perception mapping and planning in two aspects. First, our thorough comparison between the two measures confirms their significant disparity over identical perceptions. Prior studies suggested that urban planning should directly consider transforming the micro-level environment and perceptions, which are more influential on active travel behavior (Ma and Cao, 2019). Therefore, perception mapping allows us to spatially pinpoint locations requiring micro-level street improvement (He et al., 2023). Nevertheless, given the disparity found between measures for perceptions, we suggest planners not only map locations of low quality but also pay particular attention to where high perceptual differences exist. Relying solely on any measure may introduce bias, but this “low quality-high difference” approach offers an extra step that can help highlight true “priority” areas for further assessments for long-term urban management (Ye et al., 2019).

Second, the study extends the existing literature on GeoAI-based perception studies. The subjective method captures people’s impression of the visual landscape of the street environment (Kang et al., 2023), while the objective measure may be a reasonable proxy based on scholars’ interpretations of classic design theories (Ma et al., 2021). It was suggested that the final decision on choosing a measurement model could be made based on how familiar participants are with the definition of perception; an objective measure of more self-explanatory perception, such as greenness, may outperform subjective counterparts (Qiu et al., 2023). That said, when perceptions are more complicated and design-oriented, employing a panel of experts using subjective measures might be more suitable (Ewing and Handy, 2009). Our validation has indicated that non-experts might have modest consistencies when evaluating such concepts. Furthermore, the cluster analysis offers a novel perspective for jointly analyzing individual perceptions in both measures. In contrast, the FA method introduces reconstructed perception concepts that can potentially strengthen cultural identities, developing tailored transformation tactics to shape specific and inhomogeneous characteristics for disparate street environment (Ma et al., 2021). Future studies could provide actionable design guidelines for interpreting perceptions in detail and shaping preferable street environments.

Limitations

The research contains several limitations. First, the trained algorithms can integrate low-level features as explanatory variables (Chen and Biljecki, 2023) or use advanced DL models (Song et al., 2024) to improve subjective measure accuracy further. Second, alternative sampling methods can be adopted, using macroscopic a priori context (Ye et al., 2019), such as accessibility to facilities in AOIs and conducting random sampling within aggregated clusters. Third, scholars hold different opinions on the impact of demographics. Salesses et al. (2013) found no significant variation in psychological perceptions (e.g., safety) across different demographic groups, while recent studies reveal contrary results (Ramírez et al., 2021). Thus, future investigations could clarify its influences on perceptions of design qualities. Fourth, similar to the place pulse datasets (Dubey et al., 2016; Salesses et al., 2013), this study did not conduct inter-rater and inner-rater reliability tests due to time and cost constraints. Future research can increase the overlapped portions of SVI between raters to provide additional validation. Fifth, we did not measure within the variance of perceptions or use a multi-scale approach. Zhou et al. (2024) pointed out that perceptions may suffer from the multiple unit area problem depending on the unit chosen. Scholars can consider aggregating perceptions at various levels to compare their differences for more insights. Finally, our research mainly involves exploratory analysis. Future studies can follow our comparative framework to evaluate predictive power in various scenarios, scrutinizing their applicability in explaining socioeconomic statuses.

Conclusions

This study pioneers the systematic comparison of subjective and objective measures of micro-level street perceptions in Shanghai, focusing on five urban design qualities: complexity, enclosure, greenness, imageability, and walkability. The two measures of perception include (1) subjective perceptions inferred using experts' preferences adopting a GeoAI-based approach and (2) objective perceptions quantified from derived formulas.

The study reveals four critical insights. First, we found a significant difference between the two measures for identical perception concepts using statistical analysis, including Pearson correlation and paired t-tests. In general, subjective perceptions exhibited higher mean values than their objective counterparts. Second, using subjective and objective methods, the study revealed spatial heterogeneity in the distribution pattern. Notably, perceptual differences existed in all perceptions, with straightforward perceptions (e.g., greenness) exhibiting more similarities and complex perceptions (e.g., imageability) displaying more discrepancies and even demonstrating contrary distribution patterns. Third, jointly evaluating perceptions using cluster analysis achieved higher consistency in identifying areas of low quality, with less coherence in interpreting jointly good-quality areas across measures. Finally, using Factor Analysis, several new dimensions were introduced for each measure, offering novel perspectives to investigate the perception mapping of urban streetscapes.

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Ethical statement

Ethical approval

Ethical review and approval were waived for this study due to the analyzed datasets are secondary data at the aggregated level. All personal identifiers were properly anonymized so that no participant can be identified.

Informed consent

Written informed consent was waived as the analyzed datasets are secondary data at the aggregated level. No personal identifier (e.g., participants' names) was collected during the original survey, and gender and age information were properly anonymized. Only the aggregated level information (e.g., gender ratio) were kept. Therefore, no participant can be identified.

ORCID iDs

Qiwei Song  <https://orcid.org/0000-0003-2054-2766>

Meikang Li  <https://orcid.org/0000-0002-7149-8251>

Waishan Qiu  <https://orcid.org/0000-0001-6461-7243>

Data availability statement

The data presented in this study are available on request from the corresponding author.

Supplemental Material

Supplemental material for this article is available online.

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Qiwei Song is a Ph.D. Student in Architecture and a Computational Social Science Fellow at the Chinese University of Hong Kong. His research aims to utilize data-driven computational methods to comprehensively evaluate the accessibility to amenities and the quality of life in the high-density urban contexts. Previously, he obtained Master of Landscape Architecture degree from the University of Toronto and has extensive experience as a registered landscape architect and project manager, with practices in Canada and the UK. Through his studies and research, he aims to bridge the gaps between the realms of professional practice and academic enquiry.

Yuxian Fang is a master's student in urban planning at Chongqing University and urban design at the University of Sheffield. Her research interest is human behavior and built environment interaction. Yuxian's research delves into how urban spaces influence and are influenced by the people who inhabit them, aiming to create more responsive and sustainable urban environments. Yuxian has actively participated in various data-driven research and smart city projects in places in UK and China. Her academic journey is deeply rooted in her undergraduate studies in Architecture at Chongqing University.

Meikang Li is a lecturer and associate chairperson at the Department of Environmental Design of Shenzhen Technology University, teaching core landscape design and urban study studios. She is also a registered landscape architect who has years of professional practices in Canada, the USA, and China. Her projects have won many international awards, including the ASLA award, NUDA award and WLA award. Her current research focuses on urban analytics in the built environment to disclose the impact of environmental characteristics on human behaviour from socioeconomic aspects.

Jeroen van Ameijde is Assistant Professor and Director of the MSc. in Urban Design Programme at the School of Architecture, The Chinese University of Hong Kong, teaching and conducting research in architecture, urban analytics and urban design. His research focuses on the interaction between urban design, human activities and quality of life in high-density urban environments. Using computational and data-driven workflows, his work explores how public space perception, co-presence and social interaction patterns in public spaces are important factors in the development of supportive neighbourhoods and housing communities.

Waishan Qiu is an Assistant Professor in Urban Design at the University of Hong Kong. His research utilizes spatial analysis and AI to improve mobility, shareability, and sustainability. He has been involved with various data-driven research and smart city projects worldwide. Before joining HKU, he was an Associate at McKinsey & Company and he obtained a Ph.D. from Cornell University, an M.C.P. from MIT, and an M.Arch. with distinction from UCL. Waishan participates in many publications, lectures, and exhibitions and serves as reviewer for top-ranking journals.